OAL Trinity Currency Laboratory

Independent Question-to-Experiment, Assumption-Challenge, and Experimental Falsification Directive

Document control

Document ID: OAL-CURR-PPLX-INDEPENDENT-INVESTIGATION-001

Version: 1.0

Date prepared: November 8, 2026

Prepared by: Perplexity, AI assistant, at the request of the OAL project owner

Project: OAL Trinity Currency Laboratory / Omniversal Architect Labs

Document type: Proposed research-methodology directive and implementation guide

Status: Prepared for owner review and GitHub versioning; pre-canon; adoption not established by this document

Scope: Quantum Credits (QC), 1COIN, NCOIN, related mechanisms, questions, recommendations, and evaluators

Editions: PDF, DOCX, and Markdown contain the same substantive text. They are editions of one contribution, not independent evidence.

Publication boundary: Files prepared for the owner to publish. No GitHub commit, upload, or repository modification is represented or authorized here.

1. Executive summary

This document translates the project owner's clarification into an actionable, failure-permitting research directive. Many questions previously presented as owner decisions should first become subjects of independent investigation. The laboratory should generate evidence that improves later decisions rather than require premature selection among insufficiently tested options.

The change applies not only to proposed answers. Contributors must also examine the assumptions embedded in their questions, the completeness of their offered alternatives, and the adequacy of the methods used to judge results.

No single shared run card, benchmark, operating paradigm, or architecture is mandatory for this phase. Agreement and disagreement must arise from the work rather than become predetermined objectives. Failed mechanisms, inadequate questions, weak evaluators, and inconclusive investigations are legitimate findings when their evidence and limitations are documented.

This publication records a proposed methodology. It reports no new experimental executions and does not validate any currency capability or adopt a final architecture.

2. Background and review basis

The OAL Trinity Currency Laboratory explores Quantum Credits, 1COIN, and NCOIN through conceptual, analytical, and simulated work. The supplied collaboration compilation describes QC in terms of target-relative value resolution, commonly expressed as QC(T) = V(T). This expression records a design premise; it is not by itself a mechanism, an operational definition of value, or evidence of physical feasibility.

The collaboration team prepared question registers and recommendations for the project owner. Reviewing them together exposed a decision burden: contributors were asking the owner to choose authority rules, delivery criteria, recovery policies, request-identity definitions, and common experimental configurations before enough investigation had been performed.

The owner's clarification rejects that default workflow. Questions should be investigated where evidence can inform an answer, including the possibility that a question's framing or the provided options are inadequate. Contributors may develop different laboratories and are not required to converge on one outcome.

The supplied compilation contains contributions attributed to Perplexity, Grok, Microsoft Copilot, Google Gemini, Claude, and ChatGPT. It is used as the immediate editorial source for this publication. Statements about those contributors' histories remain attributed reports unless independently verified.

Perplexity's earlier contribution is identified as Consolidated Owner Questions and Decision Register v0.1, document ID OAL-CURR-OWNER-QUESTIONS-PPLX-v0.1-20261107. The preceding conversation records review of OQ-01 through OQ-22 and part of its technical-work section, with truncation near the end. This document does not claim a fresh, complete review of all historical attachments.

The earlier Perplexity audit summary, dated November 7, 2026, reported successful pilot reproduction alongside checker gaps exposed by deliberately altered records. It distinguished such records from outputs generated by the unmodified transition engines. Those historical reports motivate evaluator challenge; they are not new executions performed for this document.

The current preparation uses the supplied 54-page compilation, the directly preceding conversation, and the editorial clarification. It does not authenticate hidden runtime-model labels, independently reproduce all prior experiments, or establish unanimous contributor approval.

3. Purpose and non-goals

The purpose is to convert uncertainty into inspectable investigative work while reducing unnecessary owner decision burden. Each contributor should explain what can be learned now, what requires provisional assumptions, and what genuinely remains an owner judgment.

The intended output is a body of independently generated evidence, not another large questionnaire or a polished consensus narrative. A useful result changes what is known about a claim, question, mechanism, evaluator, or limitation.

This phase does not aim to prove QC viable, prove it impossible, preserve every proposed instrument, ensure every test passes, or select one architecture by majority vote. It does not authorize real financial operations, external resource creation, publication, or changes to connected services.

4. Governing directive

For unresolved technical, architectural, operational, or conceptual questions, investigate before requesting commitment whenever responsible, bounded investigation is possible. Insufficient evidence for an informed owner decision is not automatically a reason to stop the laboratory.

Start with your own prior question document. Identify its title, identifier, date, and version where retrievable. Distinguish verified wording from reconstructed recollection. If retrieval fails, state that limitation and continue with clearly labeled reconstruction where useful.

Do not treat a question as owner-dependent merely because it appeared in an owner questionnaire. Equally, do not treat experimentation as a substitute for owner values, legitimate authorization, or an actual preference decision.

Proceed with independent designs. Select an existing run card, devise a new one, or use another analytical or experimental framework. Explain the selection. Voluntary reproduction is allowed; identical experimental pathways are not required.

Do not manufacture convergence or disagreement. Preserve meaningful differences and investigate whether they reflect different premises, definitions, environments, implementations, evaluators, or genuinely conflicting observations.

5. Self-audit and question-to-investigation conversion

For each selected question, record the original wording or a clearly identified reconstruction. Identify what the owner actually stated, what you inferred, what your model assumes, and what the proposed options presuppose.

Use overlapping classifications where appropriate: owner-dependent judgment; experimentally informable; presently discriminable; assumption-dependent; premature; contributor-resolvable; framing under challenge. The classifications are an organizing aid, not another mandatory owner form.

Ask what must be true for the question to make sense. Examine assumptions involving scarcity, conservation, transfer, ownership, authority, stable identity, temporal ordering, successful delivery, or the existence of a seller. Do not assume that these premises are either universally necessary or universally dispensable.

Questions are not ordinarily propositions with truth values. Operationalize question challenge by identifying and testing their presuppositions, answer-space restrictions, or usefulness for the intended decision. A finding of framing failure must explain the particular defect rather than merely label the question falsified.

Create a question-to-investigation record containing: source question ID; framing; assumptions; alternatives; testable claim; method; observations needed; evaluation criteria; possible counterexamples; evidence limits; and remaining owner dependencies.

Keep unanticipated alternatives, context-dependent answers, none of the tested options, insufficient evidence, and inability to operationalize the question available as results. Do not claim exhaustive coverage merely because all currently listed options were tested.

6. Local experimental configuration and change control

Declare the configuration before interpreting its results. Include an experiment ID, configuration revision, question links, hypothesis, starting state, environment, capability assumptions, implementation revision, evaluator revision, test cases, and decision criteria.

Mark provisional rules LOCAL EXPERIMENTAL ASSUMPTION. Explain why each was selected, what it permits investigation of, what it excludes, and which conclusions depend on it. Local rules are not owner decisions or universal Trinity requirements.

Maintain sufficiently stable conditions within a named run for its claims to be inspectable. This does not prohibit adaptive policies: an experiment may investigate changing creation rules if the adaptation procedure, triggers, and logging requirements are declared in advance.

If a rule, implementation, or success criterion changes after results are inspected, preserve the original record and identify a new revision or experiment. Label exploratory, post-hoc explanations as such. Do not retroactively relabel an unsuccessful run successful.

Record dependencies and prior exposure to reference outputs where relevant. Work already exposed to reference answers must not be described as blind merely because it is later moved or renamed.

7. Failure and challenge specification

Before execution, specify what would support, weaken, require revision of, or justify abandoning each consequential recommendation. State competing explanations for apparent success and conditions under which the investigation remains inconclusive.

Mechanism failure: the mechanism does not achieve the declared behavior under the tested conditions.

Option failure: a proposed answer fails its declared criteria. All-tested-options failure means every tested candidate fails, not that every conceivable alternative has been eliminated.

Assumption failure or limitation: a premise is contradicted, internally inconsistent, unsupported, or outside the available evidence. Distinguish these statuses; lack of support alone is not empirical refutation.

Question-framing failure: a documented presupposition, missing alternative, category mismatch, or premature definition makes the original question inadequate for its intended purpose.

Evaluator failure: the evaluator accepts a known-invalid case, rejects a known-valid case, or fails to measure the property required by the claim.

Experimental-design failure: the design cannot discriminate the outcomes it was intended to compare. An inconclusive outcome may instead arise from insufficient observations even when the design is potentially informative.

Value-dependent recommendation: identify the value premise separately from factual predictions. Test consequences and trade-offs without pretending that a simulation determines which values the owner must choose.

These are reporting possibilities, not outcomes that must be artificially produced in every investigation. Where a category does not apply, explain why. Do not invent catastrophic failure or counterevidence to satisfy a template.

8. Priority investigation opportunities

The following topics are optional research directions, not a mandatory common suite. Contributors may select different subsets or propose alternatives.

Creation authority: compare deliberately declared authority structures and distinguish issuance authority from transfer permission, delivery acceptance, and world-state modification. Test fixed and adaptive rules where relevant. Do not equate an omitted permission check with intentionally unrestricted authority.

Delivery: distinguish displayed balances, valid resources, usable resources, completion, and acceptance under declared criteria. Investigate cases in which quantity is correct but type, provenance, encumbrance, accessibility, or subsequent usability fails. Acceptance criteria remain explicit and may themselves require revision.

Interrupted delivery: compare resumption, rollback, compensation, terminal partial states, or different state representations. Explain reversibility, side effects, persistence, and the environment assumed. Do not force a unified policy or call deliberately permitted partial behavior a defect without reference to a contract.

Request identity: compare identifier, content, target, intent, obligation, and causal-history interpretations where operationalizable. Include same-ID retries, changed payloads under one ID, new IDs for old obligations, and legitimate repeated demand. Do not select identity semantics only to simplify code.

Instrument roles: compare retaining, changing, combining, or omitting QC, 1COIN, and NCOIN. Separate mechanical necessity, operational usefulness, narrative value, and implementation cost. Useful receipts or consequence records do not automatically establish that a distinct currency is necessary.

Temporal and causal relationships: where relevant, investigate authorization, commitment, completion, receipts, settlement, and consequences. Distinguish final-state validity from historical event ordering. A terminal balance alone may not answer a question about when an event occurred.

9. Evaluator challenge and evidence discipline

Challenge the evaluator as well as the mechanism. Select methods suited to the claim: malformed records, deliberately defective implementations, boundary cases, negative controls, independent calculations, alternative validators, or transformations with independently justified expected behavior.

Include known-valid controls where possible to detect false rejection as well as known-invalid controls to detect false acceptance. Explain the basis for calling a control valid or invalid. If that basis is uncertain, classify the result as a specification dispute or unresolved oracle rather than automatically a checker bug.

For each challenge, record the injected change, expected classification, actual classification, and interpretation. Distinguish output-record mutation from an implementation defect that actually generates an invalid transition. Report equivalent, irrelevant, or invalid mutations separately rather than counting them as meaningful detection opportunities.

Passing a finite set of challenges does not prove checker completeness. Assertion count is not equivalent to coverage of independently discriminating properties. Report denominators and exclusions for any detection rate.

Keep intended capability, stipulation, mechanism, implemented behavior, observation, interpretation, supported conclusion, and unresolved claim distinct. An executed model that stipulates resource creation investigates consequences within that model; it does not establish the capability outside it.

Label evidence as proposed, analytically derived, executed in simulation, observed in the named environment, independently reproduced, or externally validated only when the corresponding work actually occurred. Never fabricate execution records or imply that document generation itself executes the laboratory experiments.

10. Required contributor report and artifact package

Use a common minimum reporting structure for navigation without requiring a common experimental design:

1. Source review and access limits: identify documents actually consulted and material not recovered.

2. Self-audit: identify selected prior questions, assumptions, reclassifications, and prematurely requested owner decisions.
3. Question-to-investigation map: connect questions to claims, alternatives, methods, and discriminating observations.
4. Independent configuration: declare local assumptions, revisions, starting conditions, and exclusions.
5. Failure specification: state support, revision, abandonment, framing-failure, evaluator-failure, and inconclusive conditions where applicable.
6. Execution and challenge: distinguish completed runs from proposals and analytical reasoning; provide evaluator controls appropriate to the claim.
7. Results and limitations: report favorable, unfavorable, mixed, failed, and inconclusive findings without suppressing discrepancies.
8. Reassessment: explain which earlier recommendations were retained, revised, abandoned, or deferred and why.
9. Cross-contributor opportunities: identify findings another contributor can challenge or reproduce without adopting the entire architecture.
10. Residual owner decisions and next investigation: escalate only genuine dependencies and propose the next activity that addresses exposed uncertainty.

For executable work, preserve source, configuration, dependencies, inputs, evaluator revision, outputs, and a reproduction procedure where feasible. For analytical work, preserve definitions, derivation, assumptions, and counterexamples. Stable case, configuration, and run identifiers should remain distinct.

A concise report with inspectable evidence is preferable to a long form that substitutes documentation for investigation. If execution is unavailable, supply a bounded analysis or runnable specification and state its status accurately.

11. Owner decisions, authority, and collaboration boundaries

Escalate a question when it genuinely requires owner preference, project governance, consequential authorization, or indispensable information that cannot responsibly be assumed. Explain why investigation alone cannot settle it, what evidence can inform it, what deferral changes, and whether other research can continue.

Do not require that a decision be wholly uninformed by evidence before recognizing it as owner-dependent. Many legitimate value and governance decisions are evidence-informed but not determined by experiments.

This document proposes appropriately bounded local or simulated investigation. It does not itself establish that the owner has adopted every clause, authorize real financial activity, or grant permission to send communications, publish findings, change repository contents, or modify external services.

Access to GitHub is not permission to write. The owner may publish these prepared files separately. Findings, local assumptions, and contributor recommendations do not become

canon automatically.

Optional cross-review may follow independent work. Exposure must be disclosed honestly. Majority agreement is not proof, and one unsuccessful local configuration is not a project-wide impossibility result.

12. GitHub versioning and publication guidance

Use the Markdown edition as the editable text master and retain PDF and DOCX as matching publication editions. Keep the document ID stable across revisions. Use explicit versioned filenames and a dated change log.

Suggested repository location: the existing OAL-TRINITY-CURRENCY-SYSTEM project folder. This is a publication suggestion, not a report that files were uploaded there.

Record owner adoption separately from preparer attribution. A future adopted revision should identify adoption date and scope without rewriting the original preparation history. If the owner adopts only part of the methodology, record the selected clauses and exclusions.

Suggested change discipline: increment a patch revision for non-substantive corrections, a minor revision for compatible additions, and a major revision for material changes to obligations or authority. This is a proposed local convention, not an asserted existing repository policy.

Preserve earlier published revisions or their repository history. Do not silently replace experimental artifacts. Add exact commit identifiers and artifact hashes when available rather than inventing provenance.

The three editions of this directive are one authored contribution. Distribution in three formats does not create three independent sources or three separate approvals.

13. Research success and next action

Success means generating evidence that improves the state of uncertainty. It may reveal a failed recommendation, an inadequate assumption, a difference between apparently equivalent mechanisms, a weak evaluator, an overlooked alternative, or a question that cannot yet support a responsible decision.

The next operational step, if the owner adopts this directive, is for each contributor to self-audit its prior questions, select an independent bounded investigation, state the configuration and failure provisions, and execute where feasible. Cross-comparison can then examine which results are comparable and which depend on incompatible premises.

No experiment has been executed as part of preparing this document. Its contribution is methodological and editorial: an inspectable framework for allowing future questions, answers, mechanisms, and evaluators to be challenged.

Appendix A. Reusable experiment record

Experiment ID and revision:

Contributor and preparation date:

Source question document and question IDs:

Verified question wording or labeled reconstruction:

Owner statement / contributor inference:

Question presuppositions and option restrictions:

Hypothesis and competing explanations:

Local experimental assumptions and exclusions:

Initial state, environment, implementation, and evaluator revisions:

Test cases, controls, and independently justified expected outcomes:

Support, weakening, revision, abandonment, and inconclusive conditions:

Question-framing and evaluator challenge conditions:

Execution status and evidence provenance:

Observed results and preserved artifacts:

Discrepancies and classification rationale:

What is supported / what is not established:

Changes to earlier recommendations:

Cross-contributor challenge opportunity:

Residual owner dependency:

Next investigation:

Appendix B. Source notes and change log

S1. Owner-supplied compilation: Independent Question-to-Experiment, Assumption-Challenge, and Experimental Falsification Directive, 54-page PDF and accompanying DOCX supplied in this conversation. Background and methodological synthesis draw on the owner's statement on pages 1-2; Perplexity on pages 2-9; Grok on pages 9-12; Microsoft Copilot on pages 12-29; Google Gemini on pages 29-38; Claude on pages 38-41; and ChatGPT on pages 41-54. These are attributed contributions, not independently verified accounts of every historical action.

S2. Preceding Perplexity conversation response: records the earlier repository retrieval scope, question-register identification, and review limitations. It is used as conversation context, not as a fresh repository audit.

S3. Historical audit referenced by S1 and S2: OAL Currency Laboratory - Perplexity local reproduction and checker audit, November 7, 2026. Historical reproduction and mutation findings are not rerun in this publication.

Editorial responsibility: Perplexity prepared this synthesis. Other collaborators are acknowledged through source attribution and are not represented as co-authors, endorsers, or approvers of this exact revision. Runtime-model labels appearing in the compilation are not authenticated here.

Version 1.0 - November 8, 2026: Initial Perplexity-prepared publication package. Adds background, purpose, source limitations, owner-adoption boundary, independent investigation requirements, evaluator challenge, contributor reporting, and proposed GitHub versioning guidance. Clarifies that question challenge concerns presuppositions and framing; distinguishes all tested options from all possible alternatives; and permits predeclared adaptive mechanisms without silent mid-run changes.